

# Ethan Yu

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## EDUCATION

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### Queen's University

*BASc in Mechatronics and Robotics Engineering — Dean's Scholar*

Expected May 2029

*Kingston, Ontario*

## TECHNICAL SKILLS

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**Programming Languages:** C, C++, Python, MATLAB

**Embedded Systems:** ARM Cortex-M4, AVR, Contiki OS, Raspberry Pi, SPI, UART, I2C

**Hardware & Tools:** Oscilloscope, Multimeter, Git, KiCad, Onshape, SolidWorks

## EXPERIENCE

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### Undergraduate Research Assistant

May 2026 – Aug. 2026

*Queen's MEMS Laboratory*

*Kingston, Ontario*

- Developed embedded C firmware for a battery-powered wireless sensor node on the ATmega256RFR2 running Contiki OS, using cooperative multitasking for sensing, networking, and power management.
- Authored a SPI driver for a MAX31865 RTD-to-digital converter, including register-level configuration, conversion from ADC to temperature, and fault detection/clearing for a PT1000 temperature probe.
- Designed a request/response protocol over 6LoWPAN, RPL, and UDP that decodes compact codewords from a gateway to selectively collect and transmit only the requested sensor data.
- Implemented deep-sleep power management using AVR power-save mode and a wake timer to minimize energy use between transmissions, extending estimated battery life by over 17×.

### EngLinks

Oct. 2025 – Present

*STEM and Programming Tutor*

*Kingston, Ontario*

- Provided one-on-one tutoring across first-year and upper-year engineering courses to 15+ university students.
- Guided students through assignments and exam preparation across 3 university engineering courses.

## PROJECTS

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### Modbus Sensor

Aug. 2026 – Sept. 2026

- Engineered embedded C firmware for a SAMG53-based sensor node that acquires and analyzes accelerometer data over SPI, transmitting processed measurements via Modbus RTU over RS485.
- Developed a SPI driver for a KX132-1211 3-axis accelerometer, using direct register-level PDC (DMA) control on the SPI peripheral to offload per-sample data transfer from the CPU.
- Implemented an RTC-driven acquisition scheduler collecting 9,000 samples (3,000/axis) every 2 minutes, decoupling interrupt timing from blocking acquisition to prevent peripheral stalls.

### Autonomous Rover

Jan. 2026 – Apr. 2026

- Architected Arduino code for closed-loop dual-motor speed control, using encoder interrupts to estimate wheel velocities and PID controllers to regulate PWM and direction at 100 Hz.
- Programmed Python ROS 2 nodes on a Raspberry Pi to interface with Arduino over a custom 115200-baud serial protocol, sending wheel-velocity commands and receiving encoder counts for wheel odometry and navigation.
- Implemented a 500 ms command timeout that resets wheel-speed targets and PID integrators when commands stop arriving, improving robustness to communication interruptions.

### RLC Temperature Sensor

Aug. 2026 – Sept. 2026

- Extended the open-source NanoVNA Saver software in Python with CSV export functionality, streamlining collection and analysis of frequency-response measurements from custom RLC PCBs.
- Designed and experimentally characterized multiple RLC PCB configurations with varying capacitance/inductance values to compare temperature-dependent changes in circuit bandwidth and quality factor.
- Calculated quality factor from measured resonant bandwidth across 24°C–160°C, identifying a configuration whose  $Q$  increased from 34.8 to 47.4 ( $\Delta Q = 12.6$ , approximately 36%).